English Medium

EDU TERIA

72nd BPSC

Prelims Online Batch

PHYSICS

LECTURE 07

SOUND WAVE

SOUND WAVE

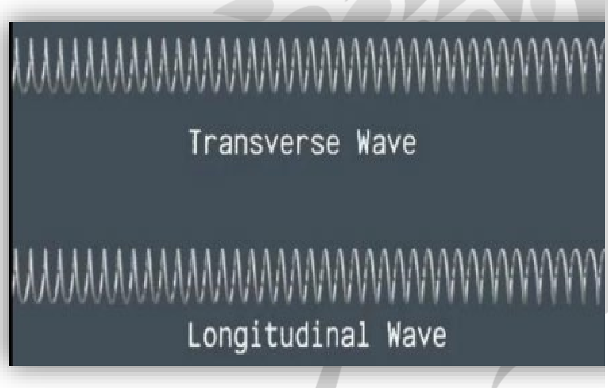
Mechanical Wave

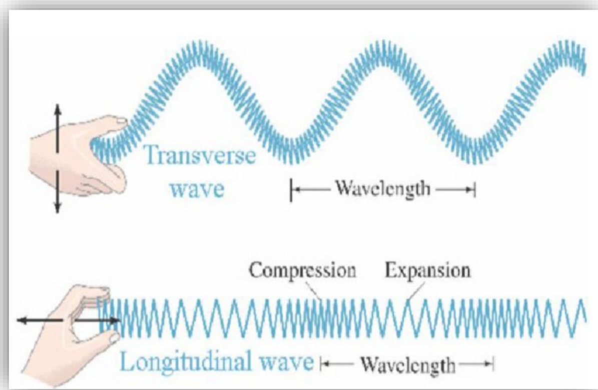
Mechanical waves are waves that require a material medium (solid, liquid, or gas) to propagate. They transfer energy without transporting matter permanently.

On the basis of oscillation of medium particle mechanical wave can be classified as—

I. **Longitudinal Waves:**
- In a longitudinal wave, the particles of the medium oscillate parallel to the direction of the wave's propagation.
- Longitudinal waves formed in air medium.
- Propagation of longitudinal/wave occurs due to change in density pressure.

II. **Transverse Waves:** The particles of the medium oscillate perpendicular to the direction of the wave's propagation.





Sound

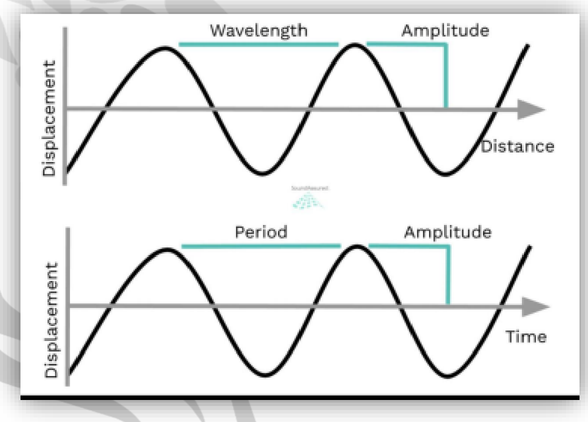
Sound is a form of energy that travels through a medium (solid, liquid, or gas) in the form of vibrations or a disturbance, which, upon reaching the ear, causes the sensation of hearing.

Key Points:

Type of Wave (Mechanical Wave): Sound is a mechanical longitudinal wave. This means the particles of the medium vibrate parallel to the direction of the wave's travel (creating high-pressure areas called compressions and low-pressure areas called rarefactions).

Medium Requirement: Sound requires a medium for propagation and cannot travel in a vacuum (like outer space).

Origin: Sound is always produced by a vibrating object.



Physical characteristics of sound wave

- **Amplitude:** The maximum displacement or pressure variation the equilibrium position. It determines the loudness of the sound higher amplitude means a louder sound.
- **Frequency:** The number of complete cycles (compressions and rarefactions) that pass a point in one second. It is measured in Hertz (Hz). Frequency determines the pitch of a sound; higher frequency equals higher pitch.

- **Wavelength:** The distance between two consecutive identical points on a wave, such as two compressions or two rarefactions.
- **Time period:** The time it takes for one complete cycle of the wave to occur. It is the inverse of frequency
- **Speed (or velocity):** The distance a sound wave travels per unit of time. It is the product of its frequency and wavelength ($v = \lambda$)
- **Intensity:** Wave Energy passing through perpendicular per unit surface area per unit time.
  - $I = \frac{P}{A}$
  - I = Intensity (W/m^2)
  - P = Power of source (Watt)
  - A = Area (m^2)

On the basis of frequency, sound wave can be classified into following —

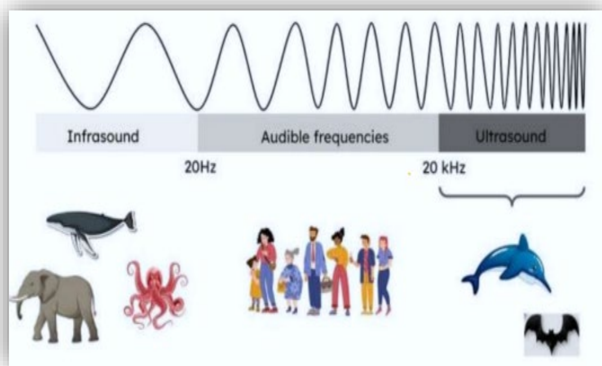
(i) **Infrasonic Sound wave**→ Range below 20Hz is produced by landslide, volcano and meteorites. Not audible to humans. Infrasonic sound waves are produced from large sources.

(ii) **(ii) Sonic sound** or **Audible sound**→ Between 20Hz to 20kHz. sensitive audible range for humans.

(iii) **Ultrasonic sound**→ Above 20 KHz frequency of sound wave known as ultrasonic sound wave can be produced and heard by some animals. **E.g.-**Bat, Cat, Dog, whale etc.

Application of ultrasonic sound waves:

1. Used in ultrasound / ultrasonography, ECG and EEG.
2. Use to kill some animal-like insects. Rat, frog etc.
3. Can be used to locate submarine objects, i.e SONAR.
4. To clean clothes and machines.



Velocity of Sound Wave:

It depends on elasticity of given medium.

Elasticity order: **Solid > Liquid > Gas**

Velocity of sound: **Solid > Liquid > Gas**

(i) Velocity of sound in solid medium

$V = \sqrt{\frac{\gamma}{\rho}}$ Y — Young's modulus of Elasity.

(ii) Velocity of sound ρ —Density of medium is liquid.

$V = \sqrt{\frac{\beta}{\rho}}$ B-Bulk modulus of elasticity

ρ- Density of liquid

(iii) Velocity of sound in air:

$V = \sqrt{\frac{P}{\rho}}$ = 319 m/s

P = pressure

ρ = Density

After LaPlace correction

Y= Degree of freedom = 1.42

$V = \sqrt{\frac{\gamma P}{\rho}}$ = 332 m/s Correct Value

$V = \sqrt{\frac{\gamma p}{\rho}} = V = \sqrt{\frac{\gamma RT}{M_w}}$

P- pressure

V- volume

m- mole

R- Gas constant

T- Temperature

R = Gas constant

M_w = Molecular weight

T = Temperature

γ – Degree of freedom-constant

Factors affecting velocity of sound:

(i) **Temperature:** On increasing the temperature medium velocity of sound increases. If temperature rises with 1°C than velocity is increased by 6 m/s. $\frac{v_1}{v_2} = \frac{\sqrt{T_1}}{\sqrt{T_2}}$

(ii) **Humidity:** On increasing humidity of air density of air decreases and velocity of sound increases.

(iii) **Pressure:** At constant temperature changes in pressure of given medium results in simultaneous change in density also. Hence, no effect on velocity of sound.

Properties of sound Save:

Reflection of Sound wave

1. Echo
2. Reverberation base on multiple Reflection of sound wave

Characteristics of sound wave:

(i) **Loudness:** Loudness decide high and low Sound.

- It is followed by intensity of sound (depends on amplitude of sound wave).
- It is measured in decibel.
- Human threshold capacity-120-140 dB without pain Loudness for Human→85 dB
- Jet – 140dB
- Normal talk 50-60 dB
- Library |Hospital– 30-40 dB

(ii) **Pitch**—9t is a sensation reserved by ear due to frequency of sound wave. Higher the frequency will result in high pitch and lower frequency considered as low pitch sound.

It measures sharpness/Shrillness of voice/sound.

(iii) **Timbre/quality**: Two sound waves having same frequency and wavelength can be of different overtone due to their different wave form.

Two sounds of some intensity and frequency can create different sensations to ear due to their different waveform.

Reverberation

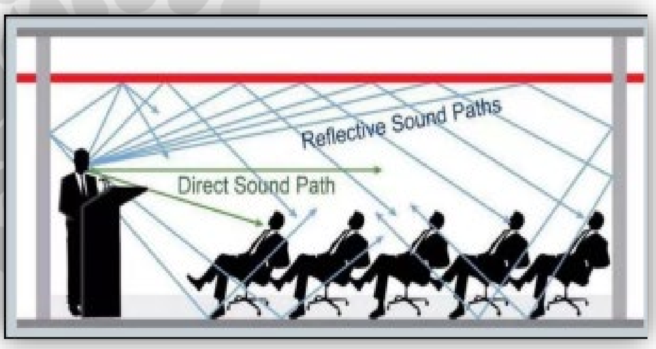
- Reverberation is the persistence of sound in an enclosed space after the original sound has stopped, caused by multiple reflections of sound waves off surfaces.

Factors that influence reverberation:

- **Size of the space:** Larger rooms tend to have more reverberation
- **Materials:** Hard, non-absorbent materials like concrete or glass cause more reflection than soft, absorbent materials like carpet or curtains.
- **Absorption:** The more sound-absorbing materials in space, the less reverberation there will be.

Measurement:

- Reverberation time (RT60) is a standard measurement for quantifying reverberation
- It is defined as the time it takes for the sound level to decrease by 60 decibels (60dB) after the sound source has stopped



Echo

An echo is the repetition of sound heard after it is reflected from a distant surface.

It occurs due to reflection of sound waves

Condition for Hearing Echo

- Minimum time gap between original and reflected sound: 0.1 second
- Based on speed of sound (~343 m/s), minimum distance:

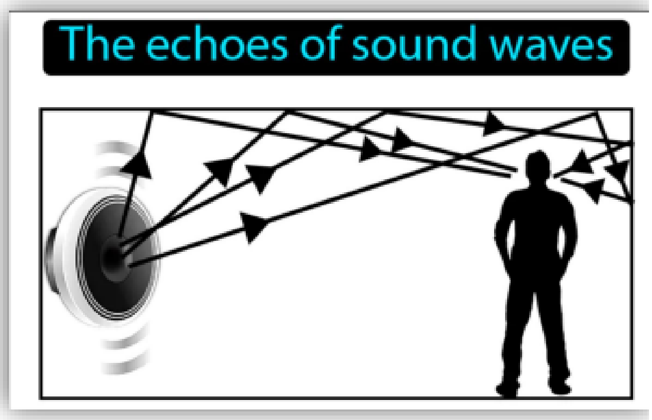
Distance ≥ 17 meters

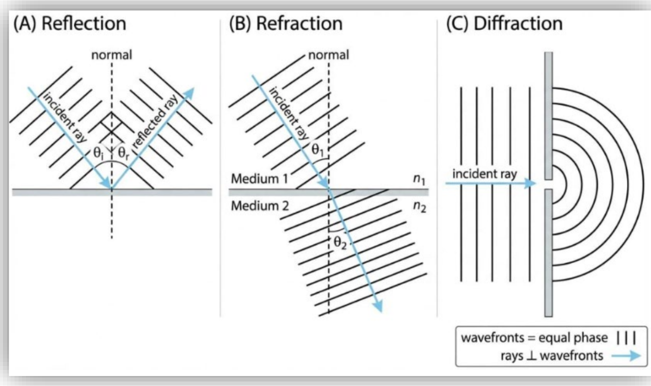
Formula of Echo

$t = 2d/v$

- t = time for echo
- d = distance of reflecting surface
- v = speed of sound

Sound travels twice the distance (to and from)





Applications of Echo

1. **SONAR** (Sound Navigation and Ranging) Used to measure depth of sea
   Detect submarines, fish
2. **Medical Imaging**
   Ultrasonic waves used in ultrasound scanning
3. **Echolocation**
   Used by bats and dolphins

Wave Phenomena

Wave phenomena are the various behaviors shown by waves when they interact with mediums, obstacles, or other waves.

Major Wave Phenomena

1. Reflection

- Bouncing back of waves from a surface
- Law: Angle of incidence = Angle of reflection
- Example: Echo (sound), mirror (light)

2. Refraction

- Bending of waves when they enter a different medium
- Caused by change in speed of wave
- Example: Sound traveling farther at night

3. Diffraction

- Bending/spreading of waves around obstacles
- Important:
- More significant when wavelength is large
- Sound shows strong diffraction

4. Interference

Superposition of two or more waves

Types:

- Constructive → Increased amplitude
- Destructive → Decreased amplitude

5. Beats

- Produced due to interference of waves of slightly different frequencies
- Beat frequency = $|f_1 - f_2|$

6. Doppler Effect

- Apparent change in frequency due to relative motion
- Examples:
- Ambulance siren
- Train horn

7. **Standing Waves (Stationary Waves)**

- Formed by superposition of two waves traveling in opposite directions

Features:

- Nodes (zero displacement)
- Antinodes (maximum displacement)

8. Polarization (Only for Transverse Waves)

- Restriction of vibrations in a particular direction

Important:

- Sound waves cannot be polarized (longitudinal nature)